



AUTHORS

Adaria Blackwell,
Hardik Chemburkar,
Ralph Dsouza

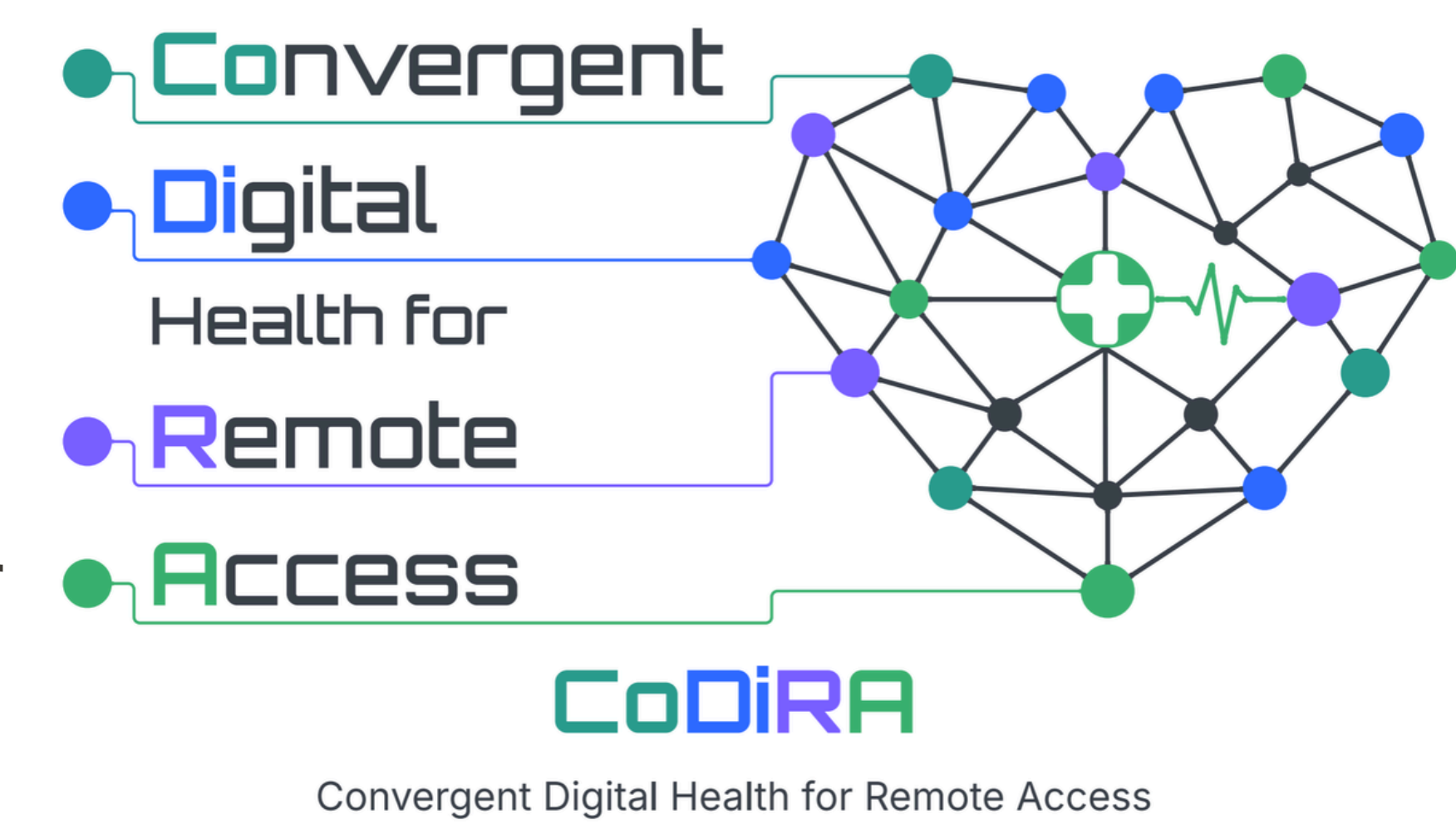
Faculty advisor: Dr. Xiao Hu



College of
Information Science

DHLitSearch: Digital Health Literacy Analysis with Semantic Search & Question Answering

Scalable NLP pipeline for semantic retrieval and question answering over digital health literature



AFFILIATIONS

University of Arizona
MS Data Science Program

01. Introduction & Motivation

- Vast amount of literature in digital health
- Transdisciplinary nature of the field
- Challenges to navigate and extract insights
- Traditional keyword search limits relevance, context, and interactivity
- Our solution: Interactive semantic literature retrieval and synthesis
 - Natural Language Processing (NLP)
 - Retrieval-Augmented Generation
 - User-centered Design
 - Supports evidence-based research and decision-making

02. Methods

- OpenAlex corpus of 414k+ digital health literacy papers
- Metadata preprocessing and duplicate removal
- 768-dimensional BioBERT embeddings for semantic retrieval
- ChromaDB vector store for similarity-based search
- RAG pipeline for source-grounded question answering
- Streamlit web app for search, Q&A, and theme exploration

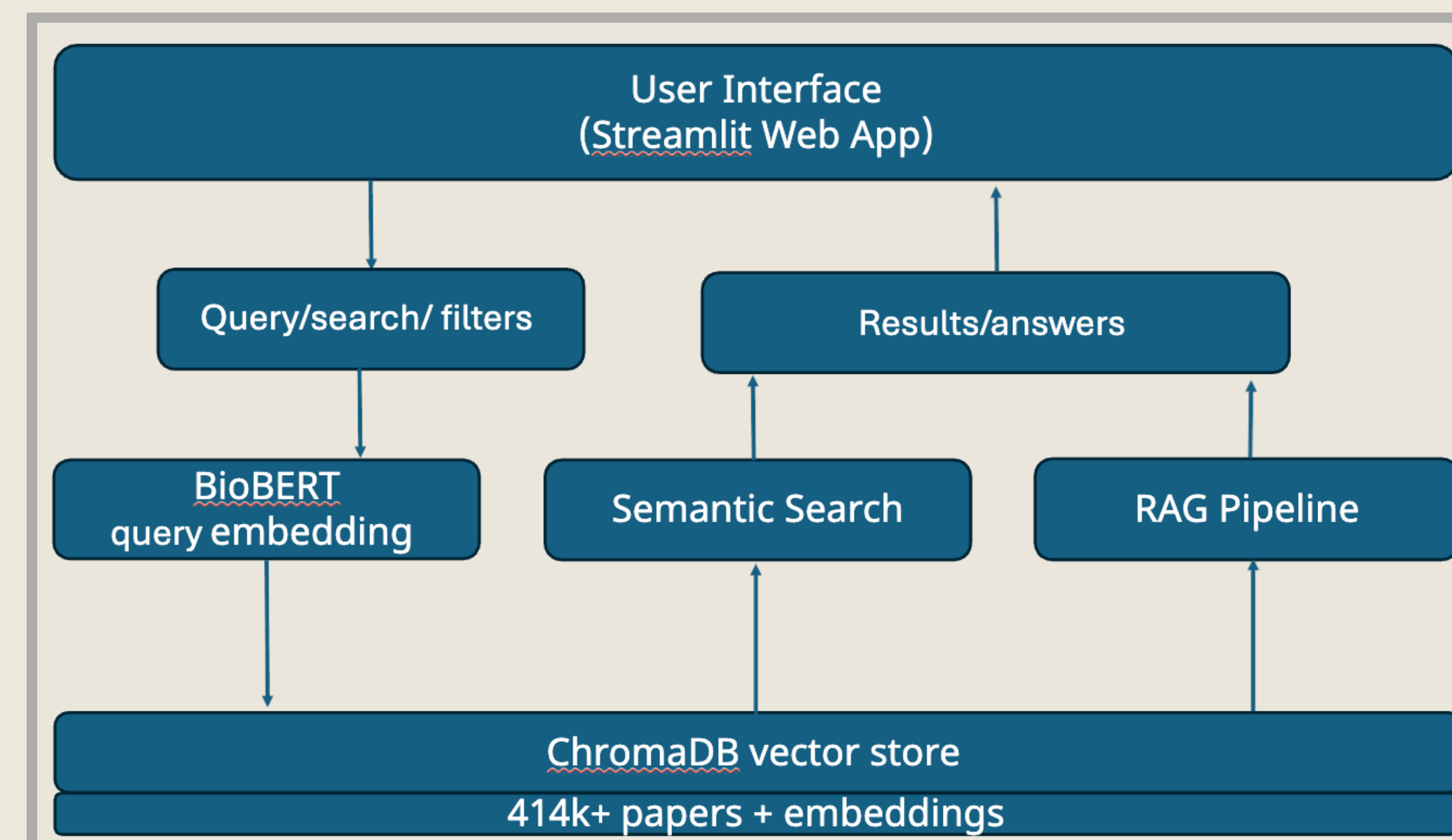


Figure 2: System architecture

03. Key Functions

- Semantic search retrieves contextually relevant papers beyond keyword matching, ranked by BioBERT cosine similarity (Fig.3)
- Top-5 paper overview synthesizes ranked abstracts into a single cited summary for fast relevance triage (Fig.3)
- RAG-based Q&A answers natural-language questions with inline DOI citations; off-topic queries are detected and refused rather than fabricated (Fig.1)
- Theme exploration clusters results into auto-labeled themes and visualizes top-keyword frequency over time (Fig.4)

DHLitSearch workspace
Ask evidence-grounded questions, search the literature semantically, and explore themes — in one place.

Q&A Search papers Themes

RAG Q&A retrieves relevant abstracts, then answers with citations linking back to the source papers. Phrase questions like you would for a colleague, e.g. "What are the main barriers to telemedicine adoption in rural areas?"

Conversation

Ask about digital health literacy, telehealth, interventions, measurement...

What is the impact of mobile health apps on diabetes self-management?

Answer Mobile health (mHealth) applications play a significant role in supporting the control and self-management of type 2 diabetes mellitus (T2DM) [Paper ID: <https://doi.org/10.2196/41076>]. These digital health interventions contribute to the management of lifestyle diseases, such as diabetes, by enhancing patient adherence, improving monitoring capabilities, and developing self-management skills [Paper ID: <https://doi.org/10.62951/ijhm.v1i2.136>]. The integration of Health Information Technology (HIT), which includes mobile health monitoring, in chronic disease management can impact patient self-management, clinical effectiveness, and quality of life [Paper ID: <https://doi.org/10.56397/jst.2024.07.07>]. Consequently, mHealth apps can empower patients to manage their conditions more effectively, potentially reducing the burden on healthcare systems [Paper ID: <https://doi.org/10.62951/ijhm.v1i2.136>].

Key findings

- Mobile health applications provide support for diabetes control and self-management in patients with Type 2 Diabetes Mellitus [Paper ID: <https://doi.org/10.2196/41076>].
- Digital health interventions, including mobile apps, improve patient adherence, monitoring capabilities, and self-management skills for lifestyle diseases like diabetes [Paper ID: <https://doi.org/10.62951/ijhm.v1i2.136>].
- The use of mobile health apps can empower patients to manage their conditions more effectively [Paper ID: <https://doi.org/10.62951/ijhm.v1i2.136>].
- Integration of Mobile Health Monitoring within Health Information Technology can impact patient self-management in chronic disease conditions [Paper ID: <https://doi.org/10.56397/jst.2024.07.07>].

Figure 1: RAG-based Q&A response with source-linked citations.

DHLitSearch
Biomedical literature search
LLM: Vertex AI

Retrieval settings
Maximum papers to retrieve: 50
Publication year range: 2010 - 2026

Research use only. Not for clinical decision-making or patient care. Verify citations in primary sources.

Recent searches
11:30 PM — health misinformation social media health literacy interventions (50 papers)

Last query: health misinformation social media health literacy interventions

Overview (top 5 papers)

Social media users widely perceive high health misinformation prevalence and difficulty in discerning it, with these perceptions correlating with sociodemographic factors, literacy, and health communication behaviors, offering insights for future interventions Gaynsynsky et al., 2024. Effective interventions include web-based programs, such as "The Appocalypse", which can promote constructive health app use, digital health literacy, and media literacy Kdnig et al., 2023, and both traditional and digital health education methods, found to improve self-efficacy, knowledge, and health literacy levels in patients with coronary heart disease Sugbarto et al., 2024. Furthermore, tailoring health literacy support is crucial, necessitating the identification of specific health literacy needs in vulnerable groups like kidney transplant recipients through profiling Siddens et al., 2024, and considering clinical characteristics alongside health and digital health literacy when optimizing the dissemination of digital health information for patients with conditions such as inflammatory bowel disease Maurud et al., 2025.

Ranked list

DOI / ID	Title	Year	Journal
https://doi.org/10.2196/51127	Perceptions of Health Misinformation on Social Media: Cross-Sectional Survey Study	2024	JMIR Infodemiology
https://doi.org/10.1111/jorc.12515	Health literacy profiles in kidney transplanted patients: A cluster analysis	2024	Journal of Renal Care
https://doi.org/10.2119/46336	The Effectiveness of Publicly Available Web-Based Interventions in Promoting Health	2023	Journal of Medical Internet Research
https://doi.org/10.1080/00365521.2025.2497952	Mapping conditional health literacy and digital health literacy in patients with inflam	2025	Scandinavian Journal of Gas
https://doi.org/10.2147/jmdh.s455431	Health Educational Methods for Improving Self-Efficacy Among Patients with Coronai	2024	Journal of Multidisciplinary H
https://doi.org/10.1111/hex.13755	Using social media listening to understand barriers to genomic medicine for those liv	2023	Health Expectations
https://doi.org/10.1177/20552076261416379	Your media diet is impacting your actual diet: The effects of influencer "What I Eat in	2026	Digital Health
https://doi.org/10.1371/journal.pone.0325202.004	The mediating effect of eHealth literacy and healthcare technology self-efficacy betw	2025	Figshare
https://doi.org/10.2196/26618	Understanding Public Perceptions of COVID-19 Contact Tracing Apps: Artificial Intell	2021	Journal of Medical Internet R
https://doi.org/10.21125/inted.2023.0251	EXAMINING THE ROLE OF EMOTIONS IN ONLINE HEALTH INFORMATION-SEEKING BEH	2023	INTED proceedings

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Figure 3: Semantic search with ranked papers (overview + citations).

Q&A Search papers Themes

Theme explorer groups a past search's papers into themes, and shows how each theme's top keywords trend over time.

Select a past search to analyze

11:30 PM — health misinformation social media health literacy interventions (50 papers)

Discover themes in this search

Top themes

Theme	Papers	Keywords
Digital Health Literacy	15	literacy, health, health literacy, digital, education, support, digital health, improving, prevention, med
Public Health Communication	12	social media, social, public, media, information, communication, health, public health, improve, heal

Smallest theme contains 10 papers (auto-chosen based on the search size).

Explore a specific theme

Select a theme

Digital Health Literacy (15 papers)

Top keywords for this theme: literacy, health, health literacy, digital, education

Papers mentioning top keywords over time — Digital Health Literacy (15 papers)

Figure 4: Theme exploration with top keywords and keyword trends.